```python
# ==========================================================
# Image Formation Model - Image Enhancement for Autonomous Driving
# Google Colab Compatible
# ==========================================================

# Install OpenCV (if not already installed)
!pip install opencv-python -q

# Import Libraries
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# ==========================================================
# Step 1: Upload an Image
# ==========================================================
print("Upload a road image (jpg/png).")

uploaded = files.upload()

# Get uploaded filename
filename = list(uploaded.keys())[0]

# ==========================================================
# Step 2: Read Image
# ==========================================================
image = cv2.imread(filename)

if image is None:
    print("Error: Unable to load image.")
else:

    # Convert BGR to RGB
    original = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

    # ======================================================
    # Low-Level Image Formation Improvements
    # ======================================================

    # 1. Noise Reduction
    denoised = cv2.GaussianBlur(original, (5,5), 0)

    # 2. Contrast Enhancement
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
    enhanced = cv2.equalizeHist(gray)

    # 3. Sharpening
    sharpening_kernel = np.array([
        [0,-1,0],
        [-1,5,-1],
        [0,-1,0]
    ])

    sharpened = cv2.filter2D(original, -1, sharpening_kernel)

    # 4. Edge Detection
    edges = cv2.Canny(enhanced, 100, 200)
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# =======================================================
# Display Results
# =======================================================

plt.figure(figsize=(15,10))

plt.subplot(2,3,1)
plt.imshow(original)
plt.title("Original Image")
plt.axis("off")

plt.subplot(2,3,2)
plt.imshow(denoised)
plt.title("Noise Reduction")
plt.axis("off")

plt.subplot(2,3,3)
plt.imshow(enhanced, cmap='gray')
plt.title("Contrast Enhancement")
plt.axis("off")

plt.subplot(2,3,4)
plt.imshow(sharpened)
plt.title("Sharpened Image")
plt.axis("off")

plt.subplot(2,3,5)
plt.imshow(edges, cmap='gray')
plt.title("Edge Detection")
plt.axis("off")

plt.tight_layout()
plt.show()

# =======================================================
# Summary
# =======================================================
print("\nImage Formation Model Improvements:")
print("✓ Noise Reduction using Gaussian Blur")
print("✓ Contrast Enhancement using Histogram Equalization")
print("✓ Image Sharpening using Convolution Filter")
print("✓ Edge Detection using Canny Operator")
print("\nThese preprocessing steps improve image quality for autonomous driving applications s
```

Upload a road image (jpg/png).

Choose Files number_plate.jpg

**number_plate.jpg**(image/jpeg) - 123440 bytes, last modified: 7/13/2026 - 100% done

Saving number_plate.jpg to number_plate.jpg


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Image Formation Model Improvements:
✓ Noise Reduction using Gaussian Blur
✓ Contrast Enhancement using Histogram Equalization
✓ Image Sharpening using Convolution Filter
✓ Edge Detection using Canny Operator

These preprocessing steps improve image quality for autonomous driving applications such as lane d
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